

Total no. of Pages:one.....

Roll no.

Degree: B-Tech Semester: 2nd

END-SEMESTER EXAMINATION, APRIL-MAY 2025

Course Code: EAEPC02

Course Title: Electronic Devices and Circuits

Max. Marks:

Duration: 03 Hours
...40.....

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain about graded and step graded semiconductor.	4	CO1
1b	Explain about varactor diode and step recovery diode.	4	CO2
1c	Explain about drift and diffusion current in detail.	4	CO1
Q 2	Attempt any 2 parts of the following		
2a	Explain Ebers Moll model for BJT.	4	CO2
2b	Draw input and output characteristics of CE npn BJT.	4	CO1
2c	BJT as a current mirror.	4	CO1
Q 3	Attempt any 2 parts of the following		
3a	Explain CMOS digital inverter.	4	CO3
3b	Explain MOSFET as CS configuration.	4	CO1
3c	When a reverse gate voltage of 15 V is applied to a JFET, the gate current is 0.001 micro A. Find the resistance between gate and source.	4	CO1
Q 4	Attempt any 2 parts of the following		
4a	Explain working principle of JFET.	4	CO1
4b	Draw V-I characteristics of JFET amplifier.	4	CO2
4c	A JFET has the following parameters : $I_{dss} = 32 \text{ mA}$; $V_{gs(off)} = -8 \text{ V}$; $V_{gs} = -4.5 \text{ V}$, Find the value of drain current.	4	CO1
Q 5	Attempt any 2 parts of the following		
5a	Write working operation of MOSFET.	4	CO2
5b	Write short notes on any two – i) LED ii) Photo diode iii) Photo transistor	4	CO3
5c	Write short notes on any two Power electronic devices – i) Thyristor ii) UJT iii) SCR	4	CO3